Name: ____________________

Class: ____________________

Date: ____________________

REVIEW — Parallel Resistors

Keep this sheet. Parallel is the mirror image of series — what was shared is now equal, and what was equal is now shared.

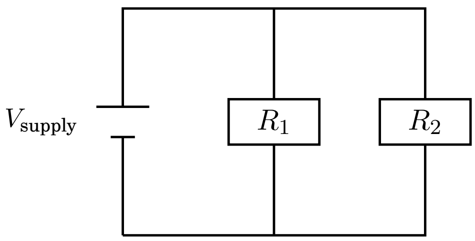
Key concept

THE THREE RULES — parallel (branches, so the charge has a choice of route)

1. **Potential difference is the same across every branch.** $V_1 = V_2 = V_{\text{supply}}$
   Each branch is connected straight across the supply, so each one gets the full p.d.
2. **Branch currents add up.** $I_{\text{total}} = I_1 + I_2$
   The current splits at the junction and rejoins after it. Nothing is lost.
3. **Total resistance is *less* than the smallest single resistor.**
   Adding a resistor in parallel always *decreases* total resistance — you have opened another route for the charge, so more current flows for the same p.d.

Mark scheme

You are NOT required to calculate the total resistance of two resistors in parallel.
(AQA Trilogy, aP2-21.) You *are* required to explain *qualitatively* why it goes down (aP2-22), and to use the two rules above. Calculations with $R_{\text{total}} = R_1 + R_2$ belong to **series** circuits only.



current splits at the junction

	Series	**Parallel**
Current	same everywhere	*splits* — branches add to the total
P.d.	*shared* between components	same across every branch
Total resistance	goes *up* $R_1 + R_2$	goes *down* — less than the smallest
Remove one	everything stops	the other branch keeps working

Worked example

WORKED EXAMPLE 1 — the standard one

A $4\,\Omega$ and a $6\,\Omega$ resistor are connected in parallel across a $12\,\text{V}$ supply. Find the current in each branch, and the total current.

1. P.d. on each branch — Each branch gets the *full* $12\,\text{V}$. (rule 1)

2. Branch 1 — $I_1 = \dfrac{V}{R_1} = \dfrac{12}{4} = 3.0\,\text{A}$

3. Branch 2 — $I_2 = \dfrac{V}{R_2} = \dfrac{12}{6} = 2.0\,\text{A}$

4. Total current — $I_{\text{total}} = I_1 + I_2 = 3.0 + 2.0 = 5.0\,\text{A}$ (rule 2)

Notice: the smaller resistor takes the *bigger* current — the opposite of what happens to p.d. in series.

REVIEW — Parallel Resistors: worked example & practice

Worked example

WORKED EXAMPLE 2 — proving rule 3 without calculating it

Using the circuit above, show that the total resistance must be less than $4\,\Omega$.

Each resistor on its own would draw less current than the pair does together. With both branches connected, 5.0 A flows out of the supply — more than the 3.0 A that the $4\,\Omega$ resistor draws by itself. Same p.d., more current, so the circuit as a whole must be offering *less* resistance than either branch alone.

You will not be asked to work out the number. You may be asked to explain exactly this.

Name: ______________________________ **Date:** ______________________

1) **State** the potential difference across each branch of a parallel circuit connected to a 6 V supply. **[1]**

2) **Calculate** the total current when two branches carry 0.30 A and 0.50 A. **[1]**

3) **Calculate** the current in each branch when a $4\,\Omega$ and a $6\,\Omega$ resistor are connected in parallel across a 12 V supply. **[4]**

4) **Calculate** the total current drawn from that supply. **[1]**

5) **State** which of those two resistors carries the larger current, and why. **[2]**

6) **Explain** qualitatively why adding another resistor in parallel *decreases* the total resistance of a circuit. **[2]**

7) Explain what happens to the second lamp when one lamp in a parallel circuit is removed, and why this differs from a series circuit. **[3]**